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ORIGINAL RESEARCH

Rain-shadow effects drive contrasting elevational patterns and assembly processes of small mammal diversity on windward and leeward slopes

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Abstract
The rain-shadow effect, a climatic phenomenon characteristic of mountain systems, creates sharply contrasting environmental conditions that shape biodiversity between the windward and leeward slopes of a mountain. Although elevational gradients in mountain biota are well studied, horizontal (slope-to-slope) diversity patterns remain comparatively unexplored. Between 2017 and 2024, we established systematic survey transects on the windward (western) and leeward (eastern) slopes of Baima Snow Mountain, Yunnan Province, China, and conducted standardized sampling of small mammals. Using the mitochondrial cytochrome-*b* gene for phylogenetic reconstruction together with traits linked to thermoregulation, osmoregulatory capacity, and resource use, we quantified taxonomic, phylogenetic, and functional diversity, identified their environmental drivers, and inferred community assembly processes. Our results revealed that leeward communities had low diversity in hot–dry valleys but markedly high diversity in high-elevation dark-coniferous forests, a pattern opposite to that observed on the windward side. Generalized linear mixed models identified humidity as the dominant predictor of diversity patterns, with temperature stability, primary productivity, and anthropogenic disturbance providing additional explanatory power. After controlling for species richness, phylogenetic and functional structure on the leeward slope shifted from clustering at low elevations to overdispersion at high elevations; the windward slope exhibited the reverse tendency. These findings underscore the context-dependent nature of biodiversity persistence and its drivers along elevational gradients. Our study refines understanding of a key protected area and offers insights applicable to mountain ecosystems worldwide, especially those shaped by comparable rain-shadow dynamics.

Keywords Community assembly · Hot–dry valley · Mountain biodiversity · Mountainside · Rain-shadow

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Introduction

Mountain regions harbour approximately 87% of the world's terrestrial vertebrate biodiversity (Rahbek et al. 2019). Clarifying how this diversity is distributed, and the mechanisms that generate and sustain it, is therefore pivotal for effective conservation and management. A defining feature of mountain biotas is the pronounced variation in diversity distribution and community composition along elevational gradients (Brown 2001; McCain and Grytnes 2010; Montaño-Centellas et al. 2019). Most studies have concentrated on these vertical patterns, evaluating hypotheses that attribute elevational diversity to spatial constraints, water–energy balance, evolutionary history, climatic stability at both short and long temporal scales, and biotic interactions, including anthropogenic disturbance (Heaney 2001; Torres-Romero and Olalla-Tárraga 2015; Wang et al. 2022).

However, mountains are inherently three-dimensional, and pronounced horizontal (slope-to-slope) differences can exist even within a single massif. A key driver of these disparities is the rain-shadow effect, which imposes sharp contrasts in temperature, precipitation, and vegetation between opposing slopes (Bhuiyan and Kogan 2010; Sklenář and Lægaard 2003). Moist air ascends the windward flank, cools, and releases precipitation; the now-dry air then descends the leeward flank, where adiabatic warming suppresses rainfall and generates a warm, desiccating foehn wind (Brinkmann 1971; Elvidge and Renfrew 2016; Stockham et al. 2018). This phenomenon occurs worldwide—from the Andes and Himalayas, and the Cascades, Sierra Nevada, and Rocky Mountains of North America (Stockham et al. 2018) and drives substantial environmental shifts, including intensified warming and aridity, vegetation turnover, and reduced primary productivity (Bhuiyan and Kogan 2010; Houston and Hartley 2003; Speirs et al. 2010; Tang 2015). Attributable to these horizontal (slope) and vertical (elevation) variations, investigations in rain-shadow regions may underscore how moisture shapes differences in diversity distribution along elevational gradients globally (Kohli et al. 2022).

Despite these pronounced abiotic contrasts, the ecological consequences of the rain-shadow dynamics, especially for animal communities, are often overlooked. Although several studies have compared differences in species composition and diversity patterns of birds and mammals across opposing slopes, few have explicitly linked these observed patterns to rain-shadow effects. Jankowski et al. (2009) showed that the rain-shadow effect produces striking differences in avian assemblages between windward and leeward slopes at identical elevations in Central America, rendering elevation alone an unreliable predictor of community composition. In addition, small mammals are underrepresented in slope-to-slope biodiversity assessments. A study in the Ailao Mountains of south-west China elucidated the general mechanisms underlying the contrasting elevational patterns of small mammal species richness between the eastern and western slopes, thereby motivating more in-depth investigations in other mountain systems (Chen et al. 2017). Onditi et al. (2022) observed that, on Mount Kenya, leeward slopes harbour lower species richness but higher phylogenetic and functional diversity than windward slopes; moreover, these diversity dimensions increase with elevation on windward slopes yet decline on leeward slopes. Collectively, such findings underscore the need to compare opposing slopes when interpreting elevational gradients and to consider multiple dimensions of biodiversity in mountain systems shaped by rain-shadow dynamics.

South-west China is recognized as a global biodiversity hotspot (Mittermeier et al. 2011). In this region, north–south-oriented mountain ranges with steep elevational relief experience a pronounced rain-shadow effect generated by warm, moisture-laden monsoons from the Indian Ocean (Pan et al. 2012; Tang 2015), making it an ideal natural laboratory for assessing rain-shadow influences on biodiversity.

Drawing on classic hypotheses of elevational diversity (Heaney 2001; McCain and Grytnes 2010), we quantified species, phylogenetic, and functional diversity of small mammals along the windward (western) and leeward (eastern) slopes of Baima Snow Mountain (BSM) in Yunnan Province, China. Phylogenetic and functional metrics were standardized by species richness to infer community-assembly processes (Keddy 1992; Webb et al. 2002). We tested two predictions (Fig. 1), (1) Under the water–energy framework (Hawkins et al. 2003a; Torres-Romero and Olalla-Tárraga 2015), we expected temperature–water gradients imposed by the rain-shadow to generate contrasting diversity patterns between slopes, with diversity at low elevations on the leeward side, subject to intense foehn conditions, being lower than on the windward side; (2) On the leeward slope, cold temperatures at high elevations and aridity at low elevations should act as environmental filters, producing more clustered phylogenetic and functional structures (Keddy 1992; Webb et al. 2002). By clarifying how vertical and horizontal gradients jointly shape small-mammal assemblages, our study informs local conservation planning and provides insights into diversity distributions in mountain systems governed by rain-shadow dynamics.

Materials and methods

Study site and field sampling

The study sites (Fig. 2) lie on the windward (western) and leeward (eastern) flanks of Zhalaqueni (5,640 m above sea level, m a.s.l.), the highest peak of BSM. This mountain

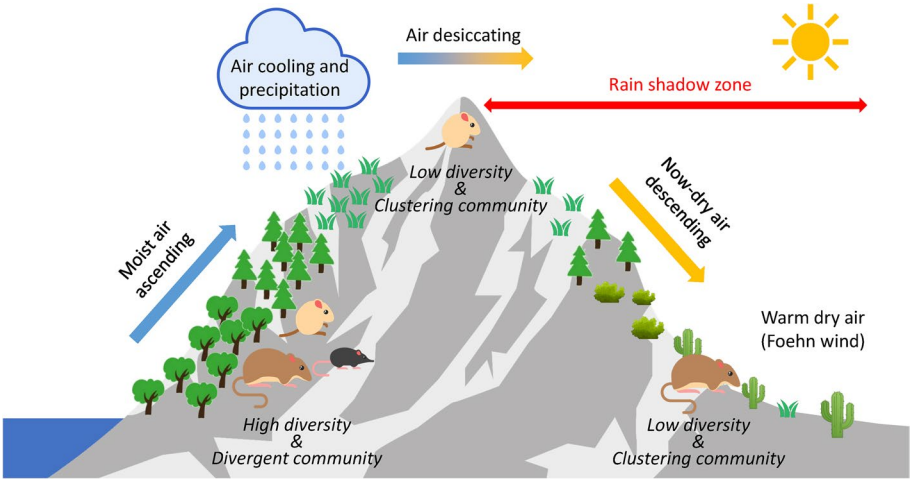


Fig. 1 Conceptional framework of this study. We hypothesize that temperature–water gradients imposed by the rain-shadow to generate contrasting small mammal diversity patterns between slopes and that phylogenetic and functional structures are more clustered in high elevations and arid low elevation

forms part of the Baimaxueshan National Nature Reserve (27°24′–28°36′ N, 98°57′–99°25′ E) within the Yunling range at the core of the Hengduan Mountains, south-western China. The Yunling chain trends north–south and intercepts the warm, moisture-laden Indian Ocean summer monsoon. Consequently, precipitation is strongly seasonal: more than 75% falls from June to September, whereas winter and spring are markedly dry (Zhang 1997). The same monsoon system produces a pronounced rain-shadow; humidity is retained on the windward slope, while the leeward slope experiences hot-dry valley conditions (Tang 2015). River valleys on both slopes lie at approximately 2,000 m a.s.l., but subsequent vegetation belts differ sharply. On the windward slope, dense subtropical shrubland occupies elevations below ~2,800 m a.s.l. (Fig. 2d); mixed broad-leaved and coniferous forest extends from ~2,800 to 4,200 m a.s.l.; *Rhododendron*-dominated shrubland occurs between 4,200 and 4,500 m a.s.l. (Fig. 2c); and above 4,500 m a.s.l. the terrain is scree-dominated, supporting only scattered alpine herbs (Fig. 2b). On the leeward slope, open semi-arid savanna bush is found below 2,800 m (Fig. 2g); continuous *Pinus* woodland occurs between ~2,800 and 3,200 m a.s.l. (Fig. 2f); humid dark coniferous forest occupies 3,200–4,200 m a.s.l.; and above ~4,200 m a.s.l. the vegetation resembles that on the windward side (Fig. 2e) (Wang 2014). Human activities are mostly concentrated in the vicinity of the river valley up to the mid-elevation pine forests on both sides of the mountain—for example, roads, croplands, and villages are distributed along the valley. Beyond these areas, the study area lies within a nature reserve, where only occasional, small-scale herding and harvesting of natural resources occur.

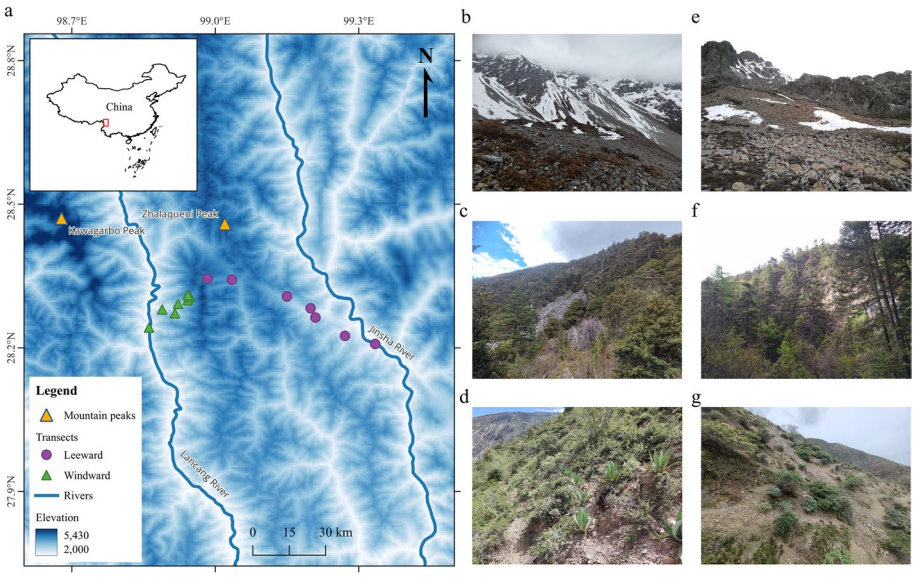


Fig. 2 Study area and the photographs depict the contrasting vegetations in river valleys between different mountain flanks, alongside forests and alpine scree habitats at mid- and high elevations. (**a**) transect established in the Baima Snow Mountain, Yunnan, South-west China; (**b**) high-elevation habitats in the windward flank; (**c**) mid-elevation habitats in the windward flank; (**d**) low-elevation habitats in the windward flank; (**e**) high-elevation habitats in the leeward flank; (**f**) mid-elevation habitats in the leeward flank; (**g**) low-elevation habitats in the leeward flank

Fieldwork was conducted at the onset of the summer monsoon during three campaigns: June 2017, July 2023, and May–June 2024. Small-mammal sampling followed the protocol of Song et al. (2020). On each mountain flank we established seven elevational transects at 400 m vertical intervals from valley floor to alpine scree, each subdivided into successive sub-transects (Fig. 2). The sub-transects were arranged consecutively to assemble a single, extended transect at each elevation gradient. Sampling along transects and sub-transects was conducted parallel to contour lines wherever feasible, maximizing coverage of habitat types within each elevational extent.

Three trap types were employed: (i) Sherman live traps (7.62 × 8.89 × 22.86 cm; H. B. Sherman Traps, Tallahassee, USA), (ii) museum snap traps (10 × 6 cm), and (iii) bucket pitfalls (Ø 20 cm × 20 cm deep). Usage of different trap types enabled us to record small mammals with different sizes and lifestyles, such as squirrels, field mice, and insectivores. Along every sub-transect we set 60 trap stations spaced 10–15 m apart. Each station comprised one Sherman trap baited with sugar-free oatmeal and one snap trap baited with fresh peanuts, positioned 1–2 m apart to minimise interference. In addition, ten bucket pitfalls were buried flush with the ground surface along putative runways within each sub-transect. Traps were deployed at dusk, inspected and re-baited at dawn, and left in situ for two consecutive nights before being relocated to the next sub-transect. Sampling continued until approximately 1,000 trap-nights (one trap × one night) had been accumulated per transect to equalise effort. Across all surveys we achieved 14,794 trap-nights, averaging ca. 1,057 per transect (range = 976–1,120).

For every captured individual we recorded body mass (BW), head-and-body length (HBL), tail length (TL), hind-foot length (HF), and ear length (EL) following Smith et al. (2010). Liver and muscle samples were excised with sterilised scissors, preserved in 99.7% ethanol in the field, and transferred to -80 °C storage on return to the laboratory. Kidneys were fixed in Bouin's solution and later stored in 75% ethanol. Voucher specimens are housed in the Vector Laboratory, Dali University, and the Kunming Institute of Zoology, Chinese Academy of Sciences.

Species captured on each transect were assembled into a site × species matrix. We calculated the sample coverage (SC) index based on the numbers of individuals recorded in each elevational transect using the 'iNEXT()' function implement in the 'iNEXT' package (Hsieh et al. 2019), setting $q = 0$ (measured as species richness). When SC exceeds 0.9, the species in the transect was considered to be adequately sampled (Ramírez-Bautista and Williams 2019). Then, abundance data were subjected to Hellinger transformation to down-weight double zeros and rendered the matrix suitable for subsequent multivariate analyses (Legendre and Gallagher 2001).

Phylogeny, traits, and phylogenetic signals

We sequenced the mitochondrial cytochrome *b* (Cyt*B*; 1,140 bp) gene to barcode each taxon and to reconstruct interspecific relationships (Tobe et al. 2010). Genomic DNA were extracted from 1 to 2 representative individuals per species with the SQ Tissue DNA Kit (D6032, Omega Bio-tek, Norcross, USA) following the manufacturer's protocol. PCR amplification and Sanger sequencing followed Koju et al. (2017) and used primer pair L14725_hsw1 (5′-ATG ACA TGA AAA ATC ATC GTT GT-3′) and H15915_hsw1 (5′-TCY CCA TTT CTG GTT TAC AAG ACC-3′). Raw chromatograms were inspected, trimmed,

and assembled in MEGA v11.0.13 (Kumar et al. 2018); contigs were aligned with the MUSCLE algorithm. The final alignment was imported into PhyloSuite v1.2.3 (Zhang et al. 2020a) for downstream analyses. PartitionFinder2 v2.1.1 (Lanfear et al. 2017), run with the greedy search strategy and Akaike's information criterion adjusted for small sample size (AICc), selected a single partition under the GTR + I + G substitution model. Maximum-likelihood phylogenies were inferred in IQ-TREE v2.2.0 (Nguyen et al. 2015) using 20,000 ultrafast bootstrap replicates (-B 20000) and the Shimodaira–Hasegawa-like approximate likelihood-ratio test (-alrt). Tree topology and nodal support values were visualised and exported in FigTree v1.4.4 (Rambaut et al. 2018). The resulting tree (Fig. S1) served as the basis for calculating phylogenetic-diversity metrics in subsequent analyses.

For phenotype traits, five external measurements including BW, HBL, TL, HF, and EL were taken in the field because they correlate strongly with thermal adaptation (Gohli and Voje 2016). Given the allometric relationships linking appendage lengths to body size, we first converted TL, HF, and EL into their respective values relative to HBL (i.e., TL/HBL, HF/HBL, and EL/HBL) of each individuals (Santoro and Calzada 2022). Then, values for exterior morphological traits were averaged by species. Kidney length, width, thickness, and the widths of the renal cortex and medulla were measured (Kohli and Rowe 2019) on preserved organs under a ZEISS Smartzoom 5 automated digital microscope. From these we derived relative medullary thickness (RMT) and percentage medullary thickness (PMT), both indices of osmoregulatory efficiency (Frean et al. 1998). Diet, diel activity pattern, and life-form (substrate use) were obtained from the COMBINE database (Soria et al. 2021) and from the comprehensive Chinese mammal trait compilation of Ding et al. (2022). Eventually, three categorical traits (diet, diel activity pattern, and life-form) and seven continuous traits (BW, HBL, TL/HBL, HF/HBL, EL/HBL, RMT, PMT) standardised by *z*-score transformation (mean = 0, SD = 1) that capture key axes of thermoregulation, osmoregulation, and resource use were included to quantify functional diversity (Table S1).

Following Mouillot et al. (2021), we converted the trait matrix to a Gower dissimilarity matrix, accommodating mixed data types. Principal Coordinates Analysis (PCoA) on this matrix produced orthogonal axes of trait variation. The first five axes (accumulating 71.12% variance in the multi-dimensional functional space) were retained for downstream functional-diversity calculations because they minimised mean absolute deviation (MAD = 0.022; Table S2) as assessed with 'quality.fspaces()' function in 'mFD' package (Magneville et al. 2021). We assessed whether the PCoA trait axes exhibited phylogenetic signal using Blomberg's K (Blomberg et al. 2003). K tests whether observed trait variance fits Brownian-motion expectations given the species tree. Values $K < 1$ indicate weaker signal (traits more labile than Brownian expectation), whereas $K > 1$ implies strong signal or niche conservatism (Grigoropoulou et al. 2022). We computed K with 1,000 randomisations of tip labels using 'multiphylosignal()' function in 'picante' package (Kembel et al. 2010); associated p-values (PIC.variance.p) were derived from the randomisation distribution, and significance was accepted at $p < 0.05$.

Diversity indices

Biodiversity was quantified with a suite of presence–absence and abundance-weighted metrics (Ricotta 2007). Taxonomic diversity was described by species richness (SR) and the Simpson diversity index (D), both calculated with 'vegan' package (v2.6-4; Oksanen et al.

2019). SR is simply the number of species recorded per sampling unit, whereas $D = 1 - \sum p_i^2$ gives the probability that two individuals drawn at random belong to different species (Simpson 1949).

Phylogenetic diversity was evaluated with Faith's phylogenetic diversity (PD; Faith 1992) and mean pairwise distance (MPD; Webb et al. 2002). PD is the sum of branch lengths connecting all species in an ultrametric phylogeny; MPD is the mean patristic distance among every species pair. Both indices were computed with 'pd()' and 'mpd()' functions in 'picante' package (v1.8-2; Kembel et al. 2010).

For functional diversity, because a few sampling sites did not meet the calculation requirements for the convex-hull-based functional richness index, we instead used the dendrogram-based functional diversity index (FD; Petchey and Gaston 2002) and distance-based Rao's quadratic entropy (Q; Botta-Dukát 2005). FD was calculated with the 'FD_dendro2()' script (https://github.com/ibartomeus/fundiv), constructed from a Gower distance matrix and UPGMA clustering (Mouchet et al. 2008). Q is the mean Gower distance between two individuals randomly selected from the community, computed with 'dbFD()' function in 'FD' package (v1.0-14; Laliberté et al. 2014).

Of the six metrics used, SR, PD and FD are incidence-based, whereas D, MPD and Q incorporate species abundances. These indices were chosen for their ecological interpretability and their mathematical relationships: MPD and Q are variants of D where species within a community possessing divergent evolutionary histories or functional traits (Botta-Dukát 2005; de Bello et al. 2016).

Explanatory variables

Because diversity distributions rarely arise from a single driver (Onditi et al. 2023; Wen et al. 2016), we quantified key environmental, productivity, and anthropogenic predictors for each transect associated with diversity patterns identified in previous literature (Bohdalková et al. 2021; Hawkins et al. 2003a; Torres-Romero and Olalla-Tárraga 2015). Nineteen bioclimatic variables describing temperature and precipitation regimes (BIO1–BIO19) spanning 1970–2000 were downloaded from WorldClim 2.1 (30″ resolution; Fick and Hijmans 2017). Atmospheric humidity was represented by the Aridity Index v3 (AI), computed as mean annual precipitation divided by potential evapotranspiration for 1970–2000, with larger values indicating wetter conditions (30″ resolution; Zomer et al. 2022). Primary productivity was measured with China's 500 m annual net primary productivity raster (2000–2024) available from the Earth Resources Data Cloud Platform (https://www.gis5g.com). Anthropogenic activity was assessed using the 2020 Human Footprint index (HFP; Mu et al. 2022), a composite of human population density, land use, infrastructure, and accessibility, provided at 1 km resolution. Because the length of the transects were approximately the grid size of the raster layers, values of each predictor were sampled at the starting coordinates of each elevational transect with the QGIS 3.38.1 "Point Sampling Tool" (QGIS Development Team 2023); extracted values are listed in Table S3.

Model fitting

We first related diversity to elevation directly by regressing each diversity metric (SR, D, PD, MPD, FD, and Q) on elevation using first-, second- and third-order polynomials. These

regression schemes indicate different potential elevational patterns of small mammal diversity: (1) linear increase or decrease; (2) unimodal peak or trough; and (3) high-altitude or low-altitude plateaus (McCain and Grytnes 2010). Models were fitted with 'lm()' function in 'stats' package, and then evaluated model parsimony with Akaike information criterion (AIC) via package 'MuMIn' v1.47.5 (Bartoń 2023). For every index the model with the lowest AIC (Table S4) was retained, and fitted values were plotted against elevation to visualise linear, unimodal or multimodal patterns.

Second, we related diversity to environmental variables while controlling for elevation. All environmental predictors were *z*-score standardised (mean = 0, SD = 1) prior to analysis so that the resulting regression coefficients were directly comparable. Pairwise Pearson product–moment correlations were calculated for the 19 WorldClim variables (BIO1–BIO19). We considered mean annual temperature to be the most accurate representation of overall temperature changes; therefore, variables exhibiting strong collinearity with BIO1 ($|r| > 0.75$) were prioritized for removal, leaving ten candidate predictors. To further limit multicollinearity, we computed variance-inflation factors (VIFs) for the candidate set and retained only variables with VIF < 5. The final environmental panel comprised six predictors: isothermality (ISO), temperature annual range (TAR), precipitation seasonality (PS), aridity index (AI), net primary productivity (NPP), and human footprint index (HFP). These factors in aggregate captured the magnitudes of the potential influences of temperature, precipitation, water availability, resource availability, and anthropogenic activity within the study area.

Relationships between each biodiversity index and the environmental predictors were assessed by treating the windward and leeward sides collectively as a single, integrated environmental gradient, while illustrating them against elevation separately for each slope to enhance the clarity of comparisons. Generalized linear mixed models (GLMMs) were fitted using Markov chain Monte Carlo (MCMC) method using the 'MCMCglmm' package (Hadfield 2010). Compared with traditional GLMMs, this approach accommodates mixed variable types and data distributions by sampling from a sequence of conditional distributions. Each diversity metric was treated as the response variable, with six standardized environmental variables included as fixed effects, while elevation was incorporated as random intercepts to account for the inherent variability in the response variable due to spatial feature differences across the study area. The priors were specified for residual variance (R) and the random effect (G) followed an inverse-Wishart distribution with parameters variance (V) equal to 1 and degree of belief (nu) equal to 0.02, as suggested in Hadfield (2010). Model chains were run for 600,000 generations with a burn-in of 100,000 and thinning intervals of 100. To evaluate the importance of individual predictor by employing the hierarchical partitioning method, we also fitted classic GLMMs using 'lmer()' function in 'lmerTest' package (Kuznetsova et al. 2017). The hierarchical partitioning of GLMMs' products was performed with the 'glmm.hp' package (Lai et al. 2022), which decomposes the marginal R^2 of each GLMM into the independent contribution of every predictor via the "average shared variance" algorithm.

To address potential spatial autocorrelations, we derived the Moran's I from the residuals of the classic GLMMs using 'moran.test()' function in 'spdep' package (Bivand and Wong 2018). We also determined the influence of spatial autocorrelations by fitting spatial lag models (SLMs) for each diversity metric using 'lagsarlm()' function in 'spatialreg' package

(Bivand and Piras 2020). Where significant autocorrelation was detected, we compared the results of the SLMs with their non-spatial counterparts based on the lowest AICc.

Quantification of assembly processes

Finally, we quantified the relative strength of the two principal community-assembly mechanisms—habitat filtering and competitive exclusion (Keddy 1992; Webb et al. 2002) using the standardised effect sizes (SESs). SES was calculated as $\mathrm{SES} = (\mathrm{observed} - \overline{x}\,null\,))\,/\mathrm{SD}\,(\mathrm{null})$, following Gotelli and Rohde (2002). Because all sampling plots lie on a single mountain (BSM), we treated the complete set of species recorded across the transect as the regional species pool. Null communities were generated with 999 iterations of the 'independent-swap' algorithm (Gotelli 2000), which preserves both species occurrence frequencies and sample species richness while assuming that any species can occupy any plot regardless of local abiotic or biotic constraints; this algorithm has been shown to maintain appropriate Type I error rates (Kembel 2009; Miller et al. 2017).

Phylogenetic null models (for PD and MPD) were produced with 'picante' package, whereas functional null models were obtained with 'FD_dendro2()' script (for FD) and 'dbFD()' function in 'FD' package (for Rao's Q), following Swenson (2014). Positive SES values indicate phylogenetic or functional overdispersion—consistent with competitive exclusion—whereas negative SES values denote clustering attributable to habitat filtering. Values near zero imply a random pattern or a joint influence of both processes (Helmus et al. 2007; Webb et al. 2002). We considered $|\mathrm{SES}| > 1.95$ as strong evidence for the corresponding assembly mechanisms (Swenson 2014).

Changes in assembly processes along the elevational gradient were assessed with ordinary-least-squares regressions, using the phylogenetic and functional SESs as response variables and elevation as the predictor (Mason et al. 2007; Montaño-Centellas et al. 2019).

All statistical analyses and figure production were carried out in R 4.4.1 (R Core Team 2021).

Results

Species composition and diversity patterns

During the field surveys we captured 807 small-mammal individuals, representing four orders, eight families, and 24 species; the overall capture rate was 5.46%. The exterior measurements of the recorded species are as follows: BW = 4.96–262.80 g, HBL = 56–195 mm, TL = 0–246.50 mm, HF = 11–49.50 mm, EL = 0–29.50 (Table S1). On the windward slope we recorded 400 individuals from four orders, seven families, and 19 species (capture rate = 5.55%), whereas on the leeward slope we recorded 407 individuals from four orders, eight families, and 21 species (capture rate = 5.37%) (Table S5). Sample coverage analysis indicated that, of the 14 transects, nine were effectively sampled (SC = 0.9–1), four exhibited marginal coverage (SC = 0.8–0.9), and only one transect at 2,800 m a.s.l. on the windward slope was incompletely sampled (SC = 0.75) (Table S6).

Elevational patterns of small-mammal diversity differed markedly between the two slopes (Fig. 3; Table S6). On the windward slope, SR, PD, and FD all peaked at ~2,800–

3,200 m a.s.l., although only SR displayed a unimodal curve. Abundance-based indices were more variable: D, MPD, and Q dipped at 2 400 m but otherwise remained relatively constant, MPD increased slightly, whereas D and Q declined at the highest elevations.

On the leeward slope, all diversity indices remained low from 2,000 to 3,400 m a.s.l., rose sharply at 3,600 m a.s.l., and then declined slightly at the highest sites. This produced a right-skewed unimodal pattern for D and a cubic relationship with elevation for the other metrics (Fig. 3; Table S6).

Drivers of diversity distribution

The SLMs provided strong evidence that SR and D were influenced by spatial autocorrelation (spatial-lag term, $p<0.05$). No spatial-lag effect was detected for PD, MPD, FD, or Q ($p>0.05$; Table S7). The non-spatial linear regression fittings for both SR (57.69 vs. 73.47) and D (50.94 vs. 64.49) had lower AICc values compared to their spatial counterparts, indicating better fit in terms of the information criterion. The Moran's *I* tests of the GLMMs' residuals were non-significant for every response variable ($p>0.05$; Table S8), indicating that any remaining spatial autocorrelation was negligible. We therefore consider the GLMM estimates reliable with respect to spatial structure.

The MCMC model fittings identified the AI, TAR, NPP, and HFP as positive predictors of most diversity metrics, whereas ISO and PS exerted consistently negative effects (Table 1). Random-effect (elevation) variances pointed to moderate elevation-related variation in SR (Standardized deviation [SD] = 0.346), PD (SD = 0.466), MPD (SD = 0.55), FD (SD = 386),

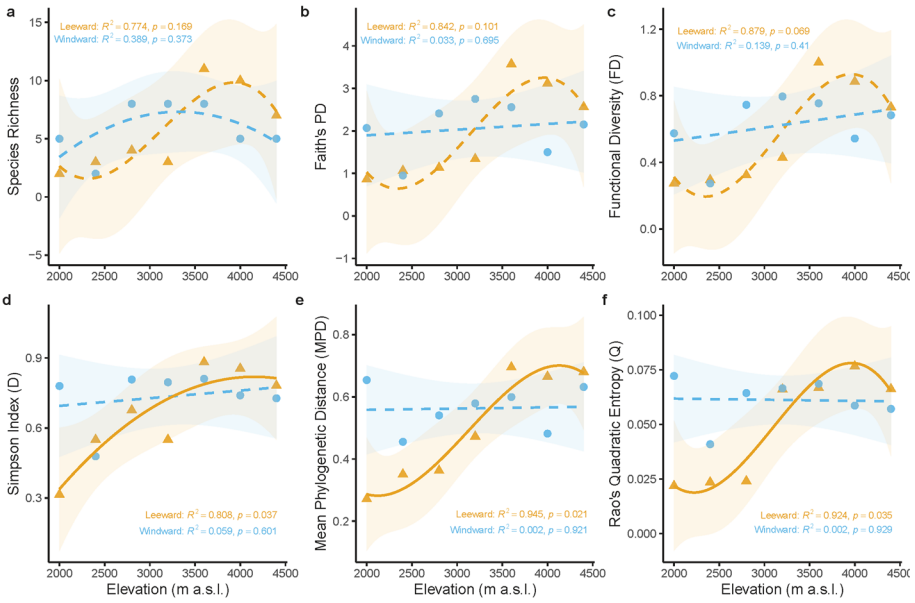


Fig. 3 Elevational distributions of taxonomic, phylogenetic, and functional diversity indices on different slopes of the Baima Snow Mountain. Fitting lines are derived from first-, second- and third-order polynomial models selected based on Akaike information criterion (AIC). The blue circle and its corresponding fitting line represent the windward (western) slope, while the orange triangle and its associated fitting line indicate the leeward (eastern) slope. Solid line is supported by strong evidence ($p<0.05$), while dash line indicating absence of statistical evidence ($p>0.05$). Shade area outlines the 95% confidential intervals

and Q (SD = 0.535), suggesting that elevation modulates these metrics, with the exception of D (SD = 0).

Hierarchical partitioning showed that AI, ISO, and TAR generally occurred as the top three individual contributors explaining the diversity metrics (Fig. 4). For most of the diversity indices (D, PD, MPD, and Q), AI emerged as the most important contributor. TAR accounted for the largest share of variance in SR, while ISO provided the largest contribution to FD, with AI ranking second and marginally lower. The influence of the remaining predictors varied according to the specific metric considered.

Null model results of phylogenetic and functional dimensions

Blomberg's *K* revealed moderate to strong phylogenetic signals in the first three PCoA axes ($K > 0.9$; $p = 0.001$) but not in the fourth ($K = 0.303$; $p = 0.601$) and fifth ($K = 0.575$; $p = 0.007$) (Table S9). Along the elevational gradients, neither phylogenetic nor functional structure displayed a consistent pattern—SES (= standardized effect sizes) alternated between positive and negative values (Fig. 5). Aspect-wise, communities on the windward slope generally showed positive and higher SES values than those on the leeward slope at low elevations, whereas the opposite trend emerged at high elevations (Fig. 5). Linear-regression results (Table S10) indicated non-significant associations between elevation and most of the SES values ($p > 0.05$), with the only significant linear correlation was found for SES. MPD on the windward slope ($p < 0.05$). Notably, on the windward slope, elevation exhibited negative estimated slopes for SES.PD, SES.MPD, and SES.Q with exception for SES.FD; by contrast, the estimated slopes were positive for all SES metrics on the leeward slope against elevation.

Discussion

Contrasting diversity patterns on different slopes

In this study, we tested two ecological predictions concerning the influences of rain-shadow effect on the slope-to-slope elevational gradients of small mammal diversity and community assembly. As anticipated, low-elevation leeward sites had lower diversity than those on the windward side. Moreover, multiple regression analysis indicated that this pattern was pri-

Table 1 Results of the Markov chain Monte Carlo generalized linear mixed models (MCMCglmms)

Diversity metric	Intercept	ISO	TAR	PS	AI	NPP	HFP	Marginal R^2
SR	-0.003	-0.447	1.601**	-1.026**	0.986**	0.842*	0.549*	0.794
D	-0.003	-0.510*	0.983**	-0.643*	1.048***	0.722*	0.338	0.777
PD	0.006	-0.521	1.636**	-1.049**	1.102**	0.857*	0.587*	0.831
MPD	0.004	-0.410**	1.085***	-0.657**	1.228***	0.792**	0.518**	0.964
FD	-0.002	-0.553*	1.560***	-1.142**	1.053**	0.709*	0.625*	0.851
Q	-0.006	-0.385	0.833**	-0.654*	1.201***	0.315	0.736*	0.843

The abbreviations for the diversity metrics are as follows: SR = species richness, D = Simpson's index, PD = Faith's phylogenetic diversity, MPD = mean pairwise distance, FD = functional diversity, Q = Rao's quadratic entropy. The explanatory variable abbreviations are as follows: ISO = isothermality, TAR = temperature annual range, PS = precipitation seasonality, AI = aridity index, NPP = net primary productivity, and HFP = human footprint index. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

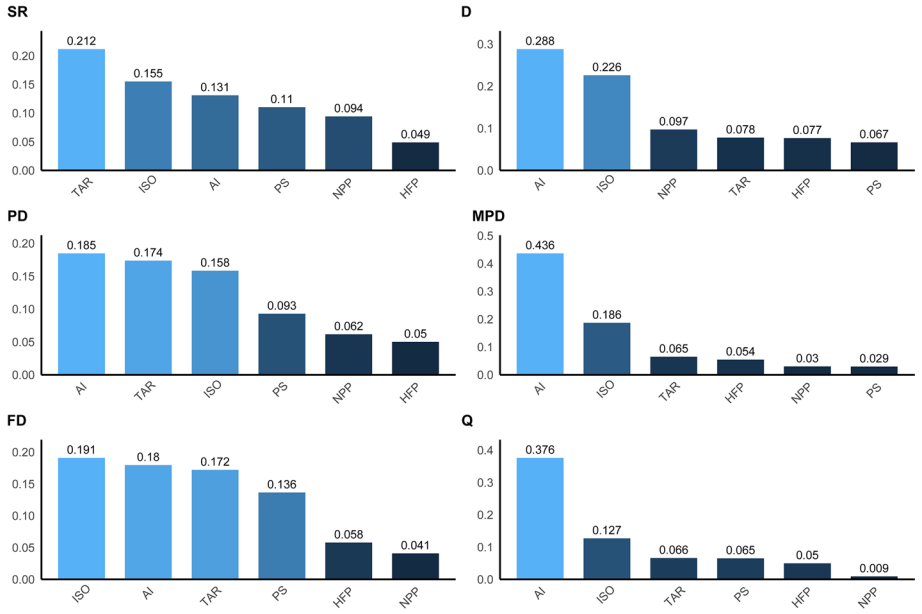


Fig. 4 Hierarchical partitioning of the individual contributions of the predicting variables of the generalized linear mixed models. The response variable abbreviations are as follows: SR = species richness, D = Simpson's index, PD = Faith's phylogenetic diversity, MPD = mean pairwise distance, FD = functional diversity, Q = Rao's quadratic entropy. The explanatory variable abbreviations are as follows: ISO = isothermality, TAR = temperature annual range, PS = precipitation seasonality, AI = aridity index, NPP = net primary productivity, and HFP = human footprint index

marily driven by water availability and climate stability, thereby supporting our first expectation. Although the vast majority of transects did not exhibit significant phylogenetic or functional overdispersion or clustering, we detected reversed trends in community assembly processes between the opposing slopes, which marginally supports our second hypothesis concerning the filtering effects of alpine harshness and valley aridity.

We found that small-mammal species composition differed markedly between the two mountain slopes. Comparable slope-related turnover along strong moisture gradients have been documented for montane birds in Central America (Jankowski et al. 2009). These results support the view that slope-specific environmental conditions, particularly moisture availability, may shape both community composition and species abundance (Musila et al. 2019).

Our investigation adhered to a standardized procedure and sampling effort was even across transects (ca. 1000 trap nights), enabling valid comparisons of diversity metrics when a few transects did not achieve complete sample coverage (Chen et al. 2017; Ramírez-Bautista and Williams 2019; Song et al. 2023). As anticipated, diversity exhibited contrasting elevational patterns between slopes of the BSM. On the windward slope, SR followed the classic unimodal relationship with elevation described for many taxa (McCain and Grytnes 2010). PD and FD likewise peaked at mid-elevations but remained relatively high at both ends of the gradient. The lack of congruence among diversity dimensions and data types indicates that comprehensive assessments should therefore integrate multiple diversity fac-

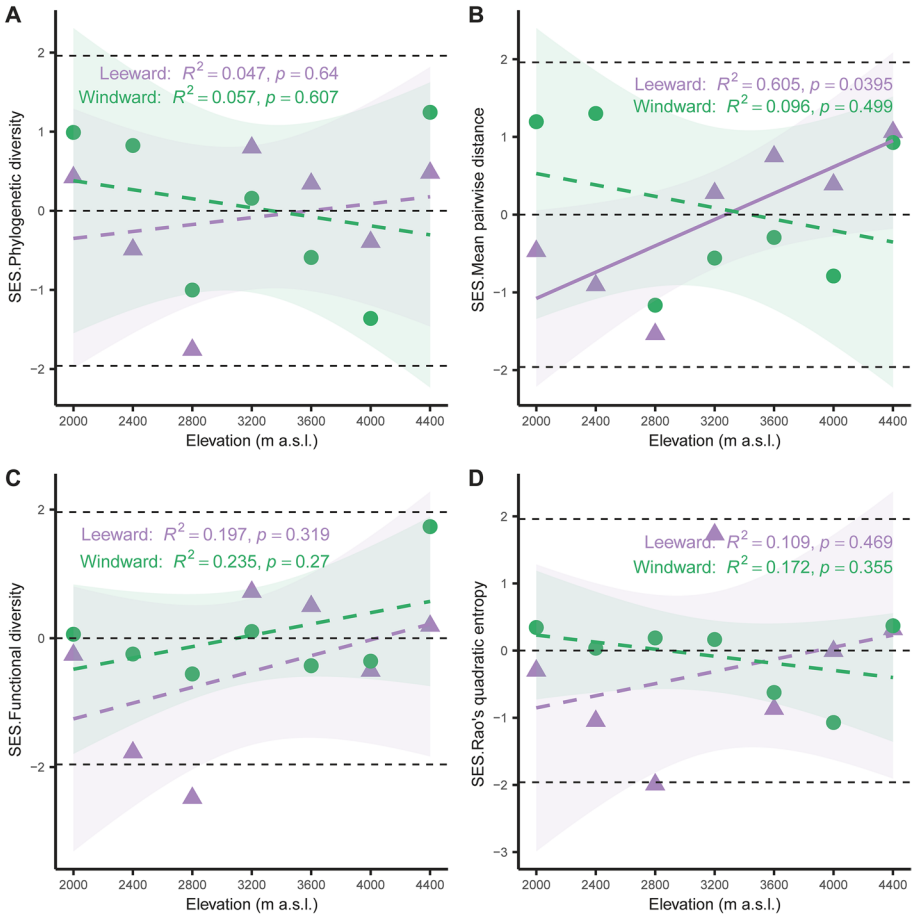


Fig. 5 Elevational patterns of the standardized effect sizes (SES) of phylogenetic and functional diversity indices based on the ordinary-least-squares regressions. The green circle and its corresponding fitting line represent the windward (western) slope, while the purple triangle and its associated fitting line indicate the leeward (eastern) slope. Solid line is supported by strong evidence ($p<0.05$), while dash line indicating absence of statistical evidence ($p>0.05$). Shade area outlines the 95% confidential intervals. Black dash lines mark the upper (1.95) and lower (-1.95) thresholds to determine strong evidence for overdispersion and clustering of phylogenetic and functional structures

ets (Ochoa-Ochoa et al. 2020; Onditi et al. 2022) and both incidence- and abundance-based data (Chalmandrier et al. 2015; Vellend et al. 2011).

All diversity metrics on the leeward slope departed from the unimodal expectation. Values remained low from 2,000 to 3,200 m a.s.l., rose sharply above 3,600 m a.s.l., and declined again in the treeless alpine scree at 4,400 m a.s.l. This pattern is consistent with our prediction that diversity at high elevations on the leeward slope benefits from monsoon-delivered moisture, whereas the low-elevation valley, subjected to intense rain-shadow and foehn conditions, supports lower diversity than the windward slope. In the adjacent Ailao Mountain, Chen et al. (2017) likewise reported a higher elevational peak of species richness on the eastern slope than on the western slope. However, they lacked assessments of

phylogenetic and functional dimensions and did not explicitly delineate how differences in ecological factors between the two slopes produced this contrast. Similar aridity-diversity gradients in multiple dimensions of small mammal diversity have been reported elsewhere (Kohli et al. 2022). Rain-shadow effects can also influence phenotype: in the shaggy soft-haired mouse, skull size is larger outside the rain-shadow (Teta et al. 2022), demonstrating that mountain rain-shadows shape both species composition and functional traits. Following Kohli et al. (2021), we also incorporated kidney measurements as a water-use trait and likewise detected slope- and elevation-dependent variation in FD. Collectively, these results indicate that multi-dimensional diversity patterns in the BSM may be a consequence of the steep environmental gradients created by the rain-shadow effect.

Magnitude of the rain-shadow effect

We identified humidity and temperature dynamics as the leading determinants of small mammal diversity in the BSM, supporting both the water–energy and climate-stability hypotheses, which predict that greater water availability and more stable temperatures can sustain larger and more diverse populations (Srivastava and Lawton 1998; Storch et al. 2018). Humidity (AI) showed a strong positive association with diversity, whereas TAR and ISO were strongly and negatively related to it. On the leeward slope, low-elevation hot-dry valleys experience semi-savanna climates, sparse shrub cover, and the widest seasonal temperature fluctuations (Elvidge and Renfrew 2016; Speirs et al. 2010; Tang 2015). By contrast, mid- to high-elevation zones remain cooler yet considerably more humid, with reduced evapotranspiration and denser woody vegetation. Globally, high temperatures are linked to elevated species richness only where water is not limiting—hot, arid regions typically harbour few species because resources are scarce (Bohdalková et al. 2021). The contrasting humidity- and temperature-diversity relationships in our study thus underscore the rain-shadow effect as a dominant ecological force shaping small mammal diversity patterns across the BSM.

Productivity (NPP) also contributed substantially to diversity patterns, consistent with the widely observed positive diversity–productivity relationship in mammals (Bohdalková et al. 2021; De Souza Ferreira Neto et al. 2022; Onditi et al. 2023). Because productivity is often tightly linked to water availability (Hawkins et al. 2003b)—a linkage evident in the BSM, where moist mid- to high-elevation forests contrast with the arid lowland valleys (Tang 2015; Wang 2014)—our findings further highlight the pivotal role of rainfall gradients generated by the rain-shadow. On the other hand, vegetation type itself could be an important factor that governing the distribution of biodiversity. However, because vegetation diversity and structural data were not well documented in the present study, we are unable to directly assess the influence of vegetation change on small mammal diversity.

Unexpectedly, the moderately positive HFP-diversity association diverged from the predominantly negative effects documented for mammals of various sizes (Onditi et al. 2023; Torres-Romero and Olalla-Tárraga 2015). At the global scale, mammal population densities have been reported to be higher in human-modified areas, which is hypothesized to result from the combined effects of species filtering, increased resources, and reduced competition and predation (Tucker et al. 2021). In our study area, two non-exclusive mechanisms may explain this pattern. First, people can act as a “human shield”, lowering predation pressure by deterring large predators and thereby boosting small-mammal persistence (Gaynor et al.

2025; Suraci et al. 2019). Second, pervasive Buddhist cultural norms that discourage harming wildlife in the region (Zou et al. 2005) likely mitigate the detrimental impacts of human presence, allowing a richer assemblage of species to coexist in human-modified habitats.

Assembly processes on vertical and horizontal directions

Although most SES metrics exhibited no significant linear relationships with elevation, the contrasting estimated slopes offer insights into community assembly processes across different sides of the mountain. In our study, habitat filtering tended to be intensified with elevation along the windward slope, conforming to existing theoretical frameworks predicting enhanced environmental harshness at higher elevations (Coyle et al. 2014; Qian et al. 2021; Zhang et al. 2020b). Conversely, the leeward slope displayed a divergent assembly pattern, corroborating our second hypothesis, which posited that habitat filtering strongly impacts community assembly at both elevational extremes (Kohli et al. 2021). This influence was particularly noticeable in the comparison between the windward and leeward valleys; the latter, characterized by hot, dry conditions and low productivity, exhibited a nuanced shift from clustering to overdispersion in community structures with increasing elevation. Although these trends did not receive strong statistical support, our findings underscore the importance of water availability as a key environmental filter in structuring small mammal diversity along mountain gradients (Kohli et al. 2022). These results prompt the notion that, due to the influence of rain-shadow effects, analogous variations in community structure are evident not only among mountains located in different regions, but even on different aspects of a single mountain massif. Considering our study's limited spatial coverage, further comparisons of biodiversity patterns between windward and leeward slopes in diverse global regions are needed to assess the generality of the present observation.

Despite that phylogenetic and functional structures displayed marked differences across slope aspects, structural changes remained internally consistent within each slope in our study. By contrast, recent investigations into mammalian (Hu et al. 2024) and avian (Ding et al. 2021) communities in the high-elevation Himalayas identified distinct patterns: phylogenetic overdispersion co-occurring with functional clustering. These patterns likely reflect evolutionary and ecological dynamics shaped by proximity to the Tibetan Plateau. In contrast to the east-west oriented Himalayas, situated at the southern boundary of the Tibetan Plateau, the BSM mountain range extends north-south into the Oriental Region. This alignment restricts east-west species dispersal due to significant river barriers, resulting in isolated, "sky island" ecosystems (He and Jiang 2014). Consequently, historical climatic fluctuations likely reinforced niche conservatism within the BSM (Li et al. 2025; Wan et al. 2018) and resulting in corresponding phylogenetic and functional structuring. Thus, our findings align with those patterns reported in other mountain ecosystems concerning rodents (Dreiss et al. 2015), bats (Cisneros et al. 2014), and birds (Lin et al. 2024) that phylogenetic and functional structures can serve as reciprocal proxies.

Conservation implications

This research underscores the combined roles of elevational gradients and associated rain-shadow effects in shaping biodiversity distributions by integrating analyses across vertical and horizontal spatial dimensions, a facet previously underexplored in ecological studies. Given that small mammals commonly serve as prey, seed dispersers, and reservoirs and vectors of pathogens, they are of critical importance for reserve management. Our study delivers critical empirical evidence that directly informs management efforts within the Baimaxueshan National Nature Reserve.

At a broader context, the rain-shadow phenomenon is prevalent in mountain ecosystems worldwide, which are frequently recognized as biodiversity hotspots (Mittermeier et al. 2011) and conservation prioritized areas (Brum et al. 2017). Traditional research focusing predominantly on elevational gradients alone have overlooked the significant variation in biodiversity patterns and underlying ecological mechanisms along horizontal gradients, even within single mountainous landscapes. The present study exemplifies a vital case study in comparable global regions, thereby indicating biodiversity conservation and management efforts should be focused at different elevations depending on which side of the mountain.

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Author contributions All authors contributed to the study conception and design. Material preparation and data collection were performed by Y-Y.L., M-R.X., G-Z.C., H-J.W., Y-H.F., X-H.S., and W-Y.S. The analysis was carried out by Y-Y.L., K.O.O., X-Y.L., and W-Y.S. The first draft of the manuscript was written by Y-Y.L. and W-Y.S. K.O.O. and X-Y.L. commented on and revised previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval Specimens and samples were acquired under the guidelines of the American Society of Mammologists on the use of wild animals. The animal studies were authorized by the internal review board of the Kunming Institute of Zoology, Chinese Academy of Sciences (Approval No: SMKX-2012023 & SMKX-20161115-01) and the Dali University Subcommittee on Experimental Animal Ethics (Approval No: 2021-P2- 162) based on the guidelines provided by the Regulations for the Administration of Affairs Concerning Experimental Animals of China. No human subjects were used in this study.

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